BLDC Controller

Technical User Manual

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Revision History

Version	Date	Author	Description
1.0	2026-07-05	Engineering Documentation Generator	Initial release

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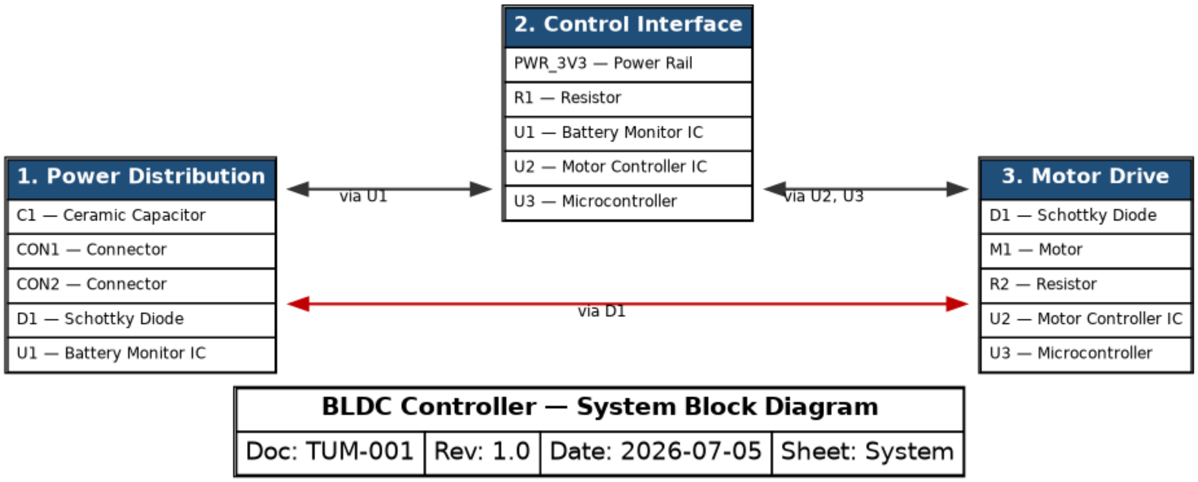
1. System Overview



The BLDC Controller is organized into 3 functional subsystems: Power Distribution, Control Interface, Motor Drive.

Power Distribution: The Power Distribution subsystem conditions and distributes electrical power to the rest of the system.

Control Interface: The Control Interface subsystem provides the digital communication backbone between the system's intelligent devices.

Motor Drive: The Motor Drive subsystem converts control commands into the phase currents that drive the motor.

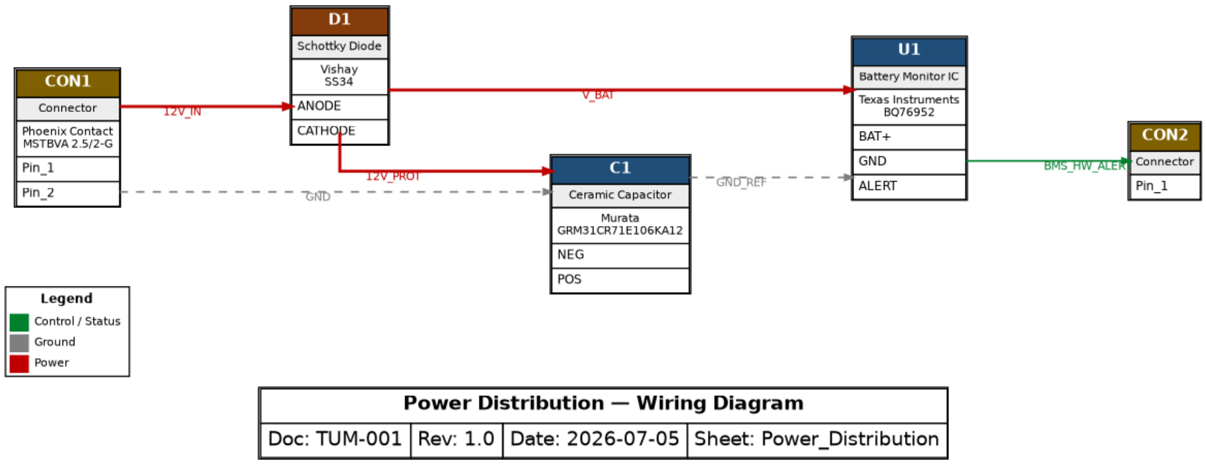
Subsystem integration: D1 spans Motor Drive and Power Distribution, carrying power signals across the boundary. U1 spans Control Interface and Power Distribution, carrying control / status, data / communication, ground, power signals across the boundary. U2 spans Control Interface and Motor Drive, carrying analog / sense, control / status, data / communication, motor phase, power signals across the boundary. U3 spans Control Interface and Motor Drive, carrying control / status, data / communication signals across the boundary.

Sub-Module Interconnections

Bridging Component	Sub-Modules	Signal Classes
D1	Motor_Drive / Power_Distribution	Power
U1	Control_Interface / Power_Distribution	Control / Status, Data / Communication, Ground, Power
U2	Control_Interface / Motor_Drive	Analog / Sense, Control / Status, Data / Communication, Motor Phase, Power
U3	Control_Interface / Motor_Drive	Control / Status, Data / Communication

2. Sub-Module Descriptions

2.1 Power Distribution



The Power Distribution subsystem conditions and distributes electrical power to the rest of the system.

Power flow: 12V_IN passes from CON1 (pin Pin_1) to D1 (pin ANODE). 12V_PROT passes from D1 (pin CATHODE) to C1 (pin POS). V_BAT passes from D1 (pin CATHODE) to U1 (pin BAT+).

Signal flow: C1 connects to U1 via GND_REF (NEG → GND). U1 signals CON2 via BMS_HW_ALERT (ALERT → Pin_1).

Role in the Overall System

Within the BLDC Controller, the Power Distribution sub-module is one of the principal functional stages. It forms the power entry and conditioning stage: every downstream function depends on the rails it establishes. It interfaces with the Control Interface sub-module through U1 (signals: BMS_ALERT, I2C_Clock, I2C_Data). It interfaces with the Motor Drive sub-module through D1 (signals: DC_BUS).

Component Roles

• C1 — Murata GRM31CR71E106KA12 (Ceramic Capacitor). 10 uF / 25 V X7R multilayer ceramic capacitor. Provides bulk decoupling and input filtering on the protected supply rail, smoothing transients before power reaches downstream ICs. In this subsystem, C1 drives GND_REF to U1; receives 12V_PROT from D1, GND from CON1.

• CON1 — Phoenix Contact MSTBVA 2.5/2-G (Connector). Phoenix Contact MSTBVA 2.5 series PCB header (5.08 mm pitch, rated 12 A / 320 V). Serves as the main power entry point for the external 12 V supply and its return. In this subsystem, CON1 drives 12V_IN to D1, GND to C1.

• CON2 (Connector). Connector used in this subsystem. In this subsystem, CON2 receives BMS_HW_ALERT from U1.

• D1 — Vishay SS34 (Schottky Diode). 3 A / 40 V Schottky barrier rectifier. Used here for reverse-polarity protection of the 12 V input: low forward drop (~0.5 V) minimizes loss while blocking reverse-connected supply voltage. In this subsystem, D1 drives 12V_PROT to C1, V_BAT to U1; receives 12V_IN from CON1.

• U1 — Texas Instruments BQ76952 (Battery Monitor IC). 16-series battery monitor and protector for Li-ion/LiFePO4 packs. Provides per-cell voltage measurement, coulomb counting, integrated protection (OV/UV/OCC/OCD/SCD), and a host interface over I2C/SPI. The ALERT output signals protection faults to the host controller. In this subsystem, U1 drives BMS_HW_ALERT to CON2; receives GND_REF from C1, V_BAT from D1.

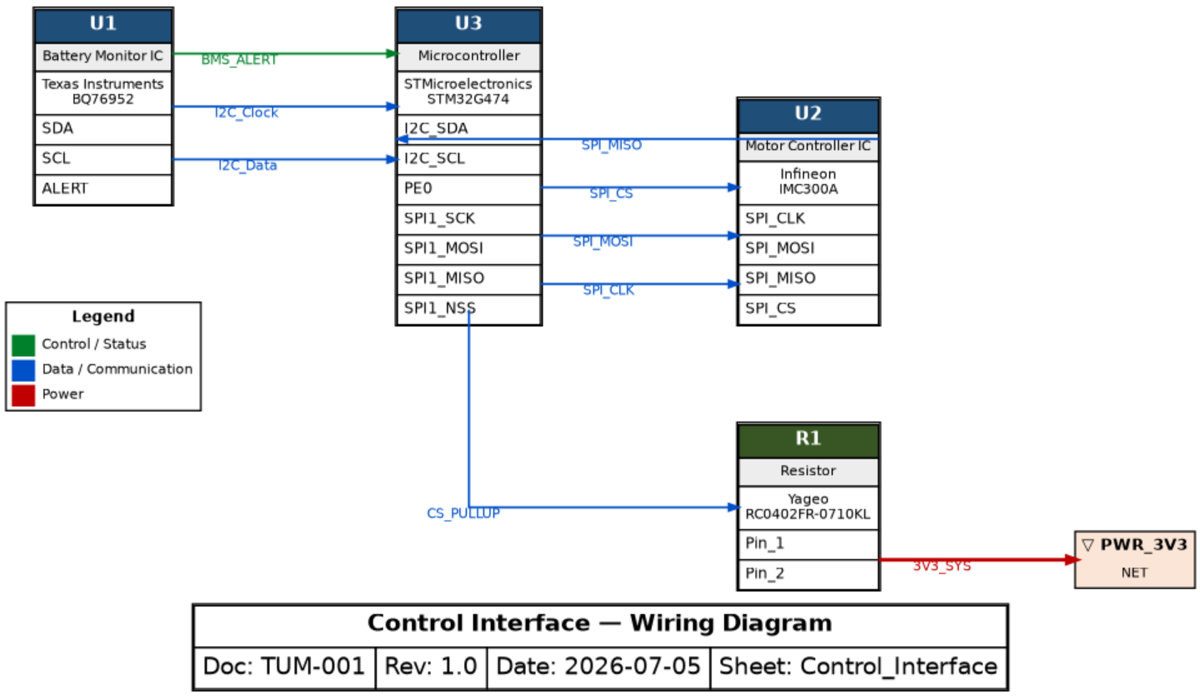
Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
Control Interface	U1	BMS_ALERT, I2C_Clock, I2C_Data
Motor Drive	D1	DC_BUS

Signal List

Signal	Class	Connections
12V_IN	Power	CON1.Pin_1 → D1.ANODE
12V_PROT	Power	D1.CATHODE → C1.POS
V_BAT	Power	D1.CATHODE → U1.BAT+
GND	Ground	CON1.Pin_2 → C1.NEG
GND_REF	Ground	C1.NEG → U1.GND
BMS_HW_ALERT	Control / Status	U1.ALERT → CON2.Pin_1

2.2 Control Interface



The Control Interface subsystem provides the digital communication backbone between the system's intelligent devices.

Power flow: 3V3_SYS passes from R1 (pin Pin_2) to PWR_3V3 (pin NET).

Signal flow: U1 communicates with U3 via I2C_Data (SDA → I2C_SDA); I2C_Clock (SCL → I2C_SCL); BMS_ALERT (ALERT → PE0). U3 communicates with U2 via SPI_CLK (SPI1_SCK → SPI_CLK); SPI_MOSI (SPI1_MOSI → SPI_MOSI); SPI_CS (SPI1_NSS → SPI_CS). U2 communicates with U3 via SPI_MISO (SPI_MISO → SPI1_MISO). U3 communicates with R1 via CS_PULLUP (SPI1_NSS → Pin_1).

Role in the Overall System

Within the BLDC Controller, the Control Interface sub-module is one of the principal functional stages. It forms the supervisory and communication stage, closing the loop between measurement, protection and actuation. It interfaces with the Motor Drive sub-module through U2, U3 (signals: DC_BUS, DRV_FAULT, PHASE_U, PHASE_V, PHASE_W, PWM_U, PWM_V, PWM_W…). It interfaces with the Power Distribution sub-module through U1 (signals: BMS_HW_ALERT, GND_REF, V_BAT).

Component Roles

• PWR_3V3 (Power Rail). Power Rail used in this subsystem. In this subsystem, PWR_3V3 receives 3V3_SYS from R1.

• R1 — Yageo RC0402FR-0710KL (Resistor). Yageo thick-film chip resistor, 0402, 1% tolerance. Used for pull-up biasing and voltage sensing dividers. In this subsystem, R1 drives 3V3_SYS to PWR_3V3; receives CS_PULLUP from U3.

• U1 — Texas Instruments BQ76952 (Battery Monitor IC). 16-series battery monitor and protector for Li-ion/LiFePO4 packs. Provides per-cell voltage measurement, coulomb counting, integrated protection (OV/UV/OCC/OCD/SCD), and a host interface over I2C/SPI. The ALERT output signals protection faults to the host controller. In this subsystem, U1 drives BMS_ALERT to U3, I2C_Clock to U3, I2C_Data to U3.

• U2 — Infineon IMC300A (Motor Controller IC). Integrated motor control IC combining a motion-control engine with a three-phase gate driver. Receives PWM commands, drives the power stage for phases U/V/W, monitors the DC bus, and reports faults (overcurrent, overtemperature) via a dedicated FAULT output. In this subsystem, U2 drives SPI_MISO to U3; receives SPI_CLK from U3, SPI_CS from U3, SPI_MOSI from U3.

• U3 — STMicroelectronics STM32G474 (Microcontroller). Arm Cortex-M4 microcontroller (170 MHz) optimized for digital power and motor control, with high-resolution timers for PWM generation, fast 12-bit ADCs, and rich connectivity (SPI, I2C, UART, CAN-FD). Acts as the central controller: it supervises the battery monitor over I2C, commands the gate driver over SPI, and generates the three-phase PWM signals. In this subsystem, U3 drives CS_PULLUP to R1, SPI_CLK to U2, SPI_CS to U2, SPI_MOSI to U2; receives BMS_ALERT from U1, I2C_Clock from U1, I2C_Data from U1, SPI_MISO from U2.

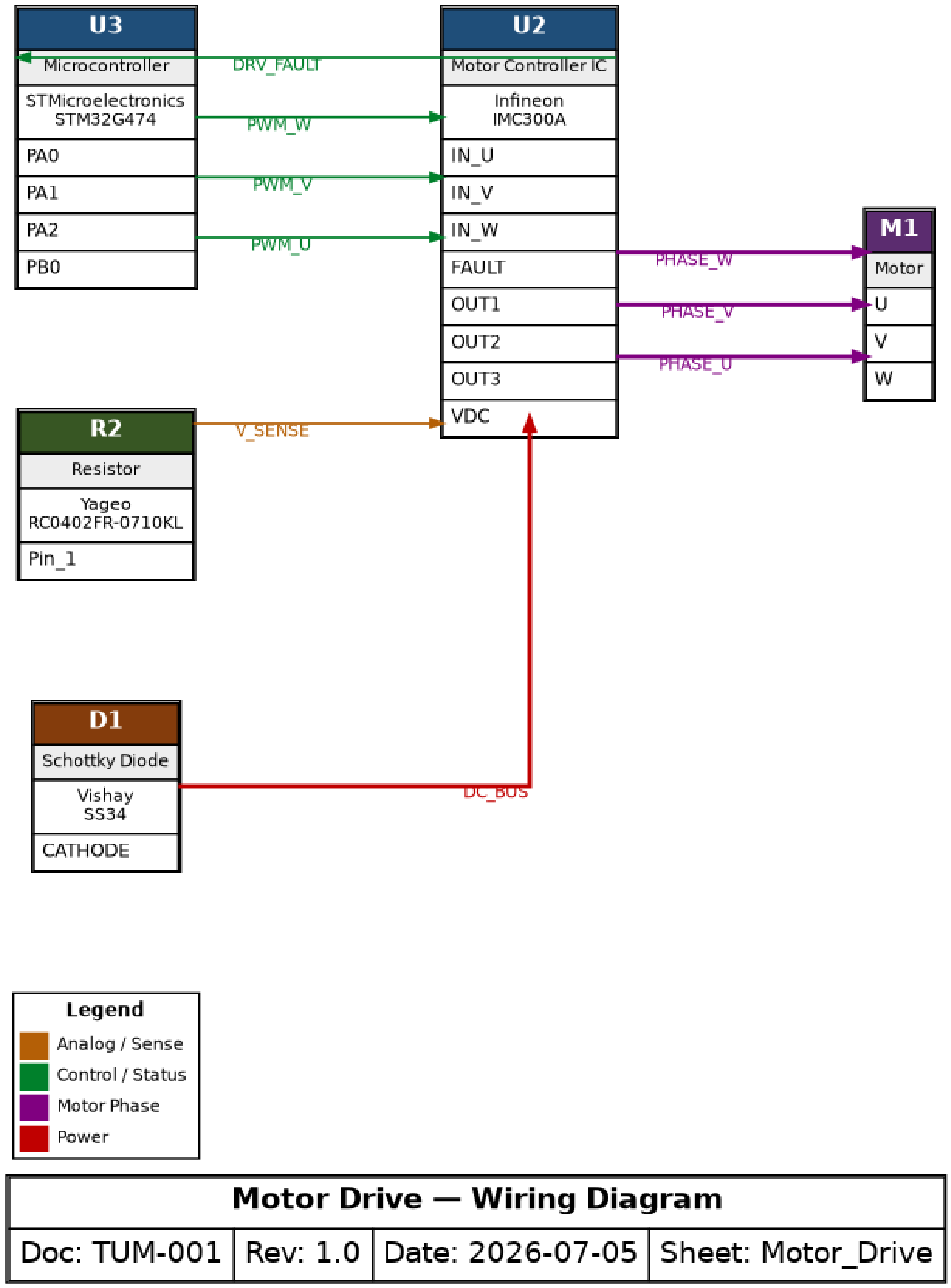
Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
Motor Drive	U2, U3	DC_BUS, DRV_FAULT, PHASE_U, PHASE_V, PHASE_W, PWM_U, PWM_V, PWM_W…
Power Distribution	U1	BMS_HW_ALERT, GND_REF, V_BAT

Signal List

Signal	Class	Connections
3V3_SYS	Power	R1.Pin_2 → PWR_3V3.NET
CS_PULLUP	Data / Communication	U3.SPI1_NSS → R1.Pin_1
I2C_Clock	Data / Communication	U1.SCL → U3.I2C_SCL
I2C_Data	Data / Communication	U1.SDA → U3.I2C_SDA
SPI_CLK	Data / Communication	U3.SPI1_SCK → U2.SPI_CLK
SPI_CS	Data / Communication	U3.SPI1_NSS → U2.SPI_CS
SPI_MISO	Data / Communication	U2.SPI_MISO → U3.SPI1_MISO
SPI_MOSI	Data / Communication	U3.SPI1_MOSI → U2.SPI_MOSI
BMS_ALERT	Control / Status	U1.ALERT → U3.PE0

2.3 Motor Drive



The Motor Drive subsystem converts control commands into the phase currents that drive the motor.

Power flow: DC_BUS passes from D1 (pin CATHODE) to U2 (pin VDC).

Signal flow: U3 signals U2 via PWM_U (PA0 → IN_U); PWM_V (PA1 → IN_V); PWM_W (PA2 → IN_W). U2 signals U3 via DRV_FAULT (FAULT → PB0). U2 drives M1 via PHASE_U (OUT1 → U); PHASE_V (OUT2 → V); PHASE_W (OUT3 → W). R2 connects to U2 via V_SENSE (Pin_1 → VDC).

Role in the Overall System

Within the BLDC Controller, the Motor Drive sub-module is one of the principal functional stages. It forms the actuation stage, converting the controller's commands into the system's physical output. It interfaces with the Control Interface sub-module through U2, U3 (signals: BMS_ALERT, CS_PULLUP, I2C_Clock, I2C_Data, SPI_CLK, SPI_CS, SPI_MISO, SPI_MOSI). It interfaces with the Power Distribution sub-module through D1 (signals: 12V_IN, 12V_PROT, V_BAT).

Component Roles

• D1 — Vishay SS34 (Schottky Diode). 3 A / 40 V Schottky barrier rectifier. Used here for reverse-polarity protection of the 12 V input: low forward drop (~0.5 V) minimizes loss while blocking reverse-connected supply voltage. In this subsystem, D1 drives DC_BUS to U2.

• M1 (Motor). Motor used in this subsystem. In this subsystem, M1 receives PHASE_U from U2, PHASE_V from U2, PHASE_W from U2.

• R2 — Yageo RC0402FR-0710KL (Resistor). Yageo thick-film chip resistor, 0402, 1% tolerance. Used for pull-up biasing and voltage sensing dividers. In this subsystem, R2 drives V_SENSE to U2.

• U2 — Infineon IMC300A (Motor Controller IC). Integrated motor control IC combining a motion-control engine with a three-phase gate driver. Receives PWM commands, drives the power stage for phases U/V/W, monitors the DC bus, and reports faults (overcurrent, overtemperature) via a dedicated FAULT output. In this subsystem, U2 drives DRV_FAULT to U3, PHASE_U to M1, PHASE_V to M1, PHASE_W to M1; receives DC_BUS from D1, PWM_U from U3, PWM_V from U3, PWM_W from U3, V_SENSE from R2.

• U3 — STMicroelectronics STM32G474 (Microcontroller). Arm Cortex-M4 microcontroller (170 MHz) optimized for digital power and motor control, with high-resolution timers for PWM generation, fast 12-bit ADCs, and rich connectivity (SPI, I2C, UART, CAN-FD). Acts as the central controller: it supervises the battery monitor over I2C, commands the gate driver over SPI, and generates the three-phase PWM signals. In this subsystem, U3 drives PWM_U to U2, PWM_V to U2, PWM_W to U2; receives DRV_FAULT from U2.

Interconnections with Other Sub-Modules

Connected Sub-Module	Via Components	Signals
Control Interface	U2, U3	BMS_ALERT, CS_PULLUP, I2C_Clock, I2C_Data, SPI_CLK, SPI_CS, SPI_MISO, SPI_MOSI
Power Distribution	D1	12V_IN, 12V_PROT, V_BAT

Signal List

Signal	Class	Connections
DC_BUS	Power	D1.CATHODE → U2.VDC
DRV_FAULT	Control / Status	U2.FAULT → U3.PB0
PWM_U	Control / Status	U3.PA0 → U2.IN_U
PWM_V	Control / Status	U3.PA1 → U2.IN_V
PWM_W	Control / Status	U3.PA2 → U2.IN_W
PHASE_U	Motor Phase	U2.OUT1 → M1.U
PHASE_V	Motor Phase	U2.OUT2 → M1.V
PHASE_W	Motor Phase	U2.OUT3 → M1.W
V_SENSE	Analog / Sense	R2.Pin_1 → U2.VDC

3. Bill of Materials

Curated bill of materials derived from the connectivity data. Entries marked 'TBD' were inferred from connectivity context and require confirmation before procurement.

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
U2	Infineon	IMC300A	IMC300A	Motor Controller IC	Integrated motor control IC combining a motion-control engine with a three-phase gate driver.	1
C1	Murata	GRM31CR71E106KA12	GRM31CR71E106KA12	Ceramic Capacitor	10 uF / 25 V X7R multilayer ceramic capacitor.	1
CON1	Phoenix Contact	MSTBVA 2.5/2-G	MSTBVA 2.5/2-G	Connector	Phoenix Contact MSTBVA 2.5 series PCB header (5.08 mm pitch, rated 12 A / 320 V).	1
U3	STMicroelectronics	STM32G474	STM32G474	Microcontroller	Arm Cortex-M4 microcontroller (170 MHz) optimized for digital power and motor control, with high-resolution timers for PWM generation, fast 12-bit ADCs, and rich connectivity (SPI, I2C, UART, CAN-FD).	1
U1	Texas Instruments	BQ76952	BQ76952	Battery Monitor IC	16-series battery monitor and protector for Li-ion/LiFePO4 packs.	1
D1	Vishay	SS34	SS34	Schottky Diode	3 A / 40 V Schottky barrier rectifier.	1
R1, R2	Yageo	RC0402FR-0710KL	RC0402FR-0710KL	Resistor	Yageo thick-film chip resistor, 0402, 1% tolerance.	2

Reference_IDs	Make	Model	Part_Number	Type	Description	Qty
CON2	TBD	TBD	TBD	Connector	Connector inferred from connectivity context — confirm part selection.	1
M1	TBD	TBD	TBD	Motor	Motor inferred from connectivity context — confirm part selection.	1

4. Connectivity Data

Control Interface

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
U1	SDA	U3	I2C_SDA	I2C_Data
U1	SCL	U3	I2C_SCL	I2C_Clock
U1	ALERT	U3	PE0	BMS_ALERT
U3	SPI1_SCK	U2	SPI_CLK	SPI_CLK
U3	SPI1_MOSI	U2	SPI_MOSI	SPI_MOSI
U2	SPI_MISO	U3	SPI1_MISO	SPI_MISO
U3	SPI1_NSS	U2	SPI_CS	SPI_CS
U3	SPI1_NSS	R1	Pin_1	CS_PULLUP
R1	Pin_2	PWR_3V3	NET	3V3_SYS

Motor Drive

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
U3	PA0	U2	IN_U	PWM_U
U3	PA1	U2	IN_V	PWM_V
U3	PA2	U2	IN_W	PWM_W
U2	FAULT	U3	PB0	DRV_FAULT
U2	OUT1	M1	U	PHASE_U
U2	OUT2	M1	V	PHASE_V
U2	OUT3	M1	W	PHASE_W
R2	Pin_1	U2	VDC	V_SENSE
D1	CATHODE	U2	VDC	DC_BUS

Power Distribution

Component_ID	Source_Pin	Target_ID	Target_Pin	Signal_Name
CON1	Pin_1	D1	ANODE	12V_IN
CON1	Pin_2	C1	NEG	GND
D1	CATHODE	C1	POS	12V_PROT
D1	CATHODE	U1	BAT+	V_BAT
C1	NEG	U1	GND	GND_REF
U1	ALERT	CON2	Pin_1	BMS_HW_ALERT